Answers: Matrix multiplication with special matrices

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Summary

Answers to questions for the study guide on matrix multiplication with special matrices.

These are the answers to Questions: Matrix multiplication with special matrices.

Please attempt the questions before reading these answers!

Q1

1.1.

$$\begin{aligned} 3x + 3y &= 4 \\ x - y &= 5 \end{aligned}$$

1.2.

$$\begin{aligned} 3x + 3y &= 4 \\ x - y &= 5 \\ -4x + 3y &= 0 \end{aligned}$$

1.3.

$$\begin{aligned} 3x + 3y + 3z &= 4 \\ y - z &= 5 \\ -4x + 3y + z &= 0 \end{aligned}$$

1.4.

$$3x + 3y + 3z = 4$$

1.5.

$$\begin{aligned} 3x + 3y + 3z + 3t &= 4 \\ x - y \qquad + t &= 5 \end{aligned}$$

Q2

2.1. $\begin{bmatrix} 9 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -4 \\ 1 \end{bmatrix}$

2.2. $\begin{bmatrix} 1 & 3 & 1 \\ -1 & -1 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ -1 \end{bmatrix}$

2.3. $\begin{bmatrix} 3 & 1 \\ 1 & -1 \\ 8 & -9 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 1 \end{bmatrix}$

2.4. $\begin{bmatrix} 3 & 3 & 3 \\ 0 & 0 & 1 \\ -4 & -4 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ 1 \end{bmatrix}$

2.5. $\begin{bmatrix} 2 & 2 & 2 & 9 \\ -1 & -1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 4 \\ -10 \end{bmatrix}$

Q3

3.1. $S0_{3\times 9} = 0_{2\times 9}$

3.2. $0_{9\times 3}T = 0_{9\times 2}$

3.3. $SI_3 = S$

3.4. $I_2 S = S$

3.5. $I_2 T$ is undefined as I_2 has two columns and T has three rows.

Q4

For 4.1 to 4.4, pick $A = B = I_2$, the 2×2 identity matrix. In this case, $AB = I_2$ which is both upper triangular and lower triangular, a diagonal matrix, and the 2×2 identity matrix.

For 4.5, pick

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \qquad \text{and} \qquad B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

where A is upper triangular and B is lower triangular. But

$$AB = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

which is neither upper nor lower triangular.

Version history

v1.0: initial version created 05/26 by tdhc.